



Name: Cohort/Batch: Date: Score:

HBASE PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is HBase and what are its core strengths? **Threat Model**

Q6 How do you monitor and tune slow queries in HBase? **Observability**

Q2 How does HBase guarantee consistency and handle transactions? **Architecture**

Q7 What backup and restore strategies does HBase support? **Recycle**

Q3 What index types does HBase support and when do you use each? **Migration**

Q8 How do you handle schema migrations in HBase? **Compliance**

Q4 How do you design a schema or data model in HBase? **Hardening**

Q9 What security features does HBase provide? **Testing**

Q5 How does HBase scale horizontally? **SSO / IdP**

Q10 What are common anti-patterns when using HBase in production? **Tuning**



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 HBase is a relational, document, key-value, graph,

Mitigation

HBase is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A6 HBase provides a slow query log, explain/explain

Observability

HBase provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A2 HBase provides either full ACID transactions or

Primitives

HBase provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A7 HBase supports logical backups (export SQL or

Automation

HBase supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A3 HBase offers B-tree, hash, full-text, spatial, or

Hardening

HBase offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A8 Schema migrations in HBase are managed with

Governance

Schema migrations in HBase are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A4 Schema design in HBase involves normalizing relations

Gotchas

Schema design in HBase involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A9 HBase supports role-based access control, encrypted

Assessment

HBase supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A5 HBase scales via replication (read replicas) for

Federation

HBase scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some HBase implementations handle this automatically; others require manual configuration.

A10 Common mistakes include selecting all columns instead

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.